6 Results, validation and discussion

We first examine the square in detail, using Section 4's solver results and additional square-only refinement and load-sensitivity checks with the same implementation [1]. We then report the comparisons across objects, loads and supports designed in Section 5, computed in the accompanying notebook.

6.1 Deformation under mesh refinement

We keep Section 4's material, full left clamp and uniform right-edge traction fixed. Table 1 compares structured meshes with n cells per side, retaining the same diagonal in every cell. These are separate from Section 5's proposed Triangle meshes. Each physical measurement uses a checked sparse-LU solution.

Table 1: New square-only refinement results from `generate_evidence.py`. Here $\bar{u}_x$ is the right-edge horizontal displacement averaged by edge length; u_x^m, u_y^m are the components at the fixed midpoint $(1, 0.5)$ m. Displacements are in mm. $\kappa_2(A)$ is estimated from checked extreme eigenpairs.

n	Free unknowns	$\bar{u}_x$	u_x^m	u_y^m	$\kappa_2(A)$
5	60	0.491352	0.489202	0.010710	6.72×10^2
10	220	0.492877	0.490810	0.003769	2.47×10^3
20	840	0.493497	0.491599	0.001255	9.10×10^3
40	3280	0.493744	0.491900	0.000418	3.45×10^4

The mean extension increases towards about 0.494 mm, while its successive changes shrink from 0.00153 to 0.000620 and 0.000247 mm. The midpoint's vertical movement also decreases. We integrate the edge trace rather than average nodes, and interpolate at the fixed midpoint when it is not a node. Absolute differences avoid dividing by its small vertical component. Every reported mesh difference exceeds the summed algebraic uncertainty estimates from the two solutions by more than 10^6. Those estimates use the actual observation weights, residuals and smallest eigenvalues; they are not certified bounds. The trend supports reduced mesh dependence of these displacements, not a known continuum error.

6.2 Solver behaviour and conditioning

Section 4's recorded timings show why iteration count alone is not a speed ranking. Diagonal PCG used 226 steps against CG's 242, but their minimum setup-inclusive times were 4.99 and 4.31 ms, respectively. Full GMRES used 228 steps yet took 146 ms. These measurements follow Section 3's protocol; small timing differences do not establish a general ranking.

Meanwhile, conditioning worsens under refinement despite the settling displacements. At $n = 20$, symmetric diagonal scaling $P^{-1/2}AP^{-1/2}$ reduces the estimated condition number from 9104 to 8029, where $P = \mathrm{diag}(A)$ as in Section 3. This modest improvement does not guarantee faster PCG. The spectral diagnostics also explain the failures. Ordinary Jacobi's iteration matrix has spectral radius about 1.96, meaning that some error components grow. Damping by 0.5 reduces this to about 0.999816, below one but close enough to predict slow asymptotic decay [2]. This agrees with the recorded divergence of Jacobi and budget exhaustion of damped Jacobi. Steepest descent also exhausted the budget. Their runtimes are not times to the requested accuracy. Among accepted runs, the largest relative difference from the same-mesh LU reference was 1.74×10^{-7}, for Gauss–Seidel, despite its residual being below 10^{-8}. Residual and displacement error are different quantities.

6.3 Sensitivity to small load changes

At the fixed 20-cell mesh, we compare two load changes of equal relative size, $\|\delta b\|_2/\|b\|_2 = 10^{-3}$. The first scales the original load. The second redistributes horizontal traction along the right edge, with its increment proportional to $2y/L - 1$, where $L = 1$ m is the edge height. Its integrated added force is zero, but its moment changes. The coefficient is chosen to give the stated load-vector norm; edge forces are integrated consistently.

We solve both changed systems with the same matrix and measure

$$G = \frac{\|\delta x\|_2/\|x\|_2}{\|\delta b\|_2/\|b\|_2}, \tag{1}$$

where x is the baseline free-displacement vector and δx its change. Uniform scaling gives $G \approx 1$, as linearity predicts, while redistribution gives $G \approx 1.77$. Both are far below $\kappa_2(A) \approx 9104$: conditioning describes possible amplification, not what every load direction produces. Neither test searches for the worst case. The Euclidean norm here compares free nodal components on one fixed mesh, not displacement fields on different meshes.

6.4 Other objects, loads and supports

Table 2 compares the three objects of Section 5 at the middle mesh-area limit (0.00125 m^2). All 36 cases passed every acceptance check of Section 5.

Table 2: Objects under the baseline clamp and traction, middle mesh. $\bar{u}_x$ is the mean right-edge horizontal displacement and u_x^m the horizontal displacement at the right-edge midpoint, both in mm. CG and PCG give iterations to a relative residual of 10^{-8}.

Object	Free unknowns	$\bar{u}_x$	u_x^m	$\kappa_2(A)$	CG	PCG
Square	1282	0.4936	0.4918	1.12×10^4	268	245
Rectangle 2×1	2532	0.9936	0.9937	1.01×10^5	427	395
Square with opening	1246	0.5560	0.5883	1.30×10^4	283	256

Doubling the length doubles the mean extension (a factor of 2.01), as expected for a plate in nearly uniform tension. The condition number grows about nine times and CG needs 427 instead of 268 iterations. The opening removes 4% of the material but increases the mean extension by 13% and the midpoint extension by 20%, because the load must flow around it; the condition number rises by only 16%.

Under refinement, the mean extension of the square and the rectangle changes by less than 0.04% at the last step, but that of the perforated square changes by 0.66%. The perforated square is therefore the most mesh dependent, consistent with stresses concentrating at the corners of the opening. All successive differences exceed the algebraic uncertainty estimates.

The displacements scaled exactly with the load magnitude. Moving the same total force to the lower or upper quarter of the right edge leaves the mean extension almost unchanged (0.4949 against 0.4920 mm for the middle patch on the square), but bends the object: the right-edge midpoint moves vertically by ± 1.07 mm on the square and ± 4.42 mm on the rectangle. Clamping only the central half of the left edge increases the mean extension by 22% (square), 11% (rectangle) and 25% (perforated square), and multiplies $\kappa_2(A)$ by 2.4 to 3.0.

The solver comparison repeats the square's pattern on every object. CG, PCG, MINRES and GMRES reached the target on all 27 distinct systems. Jacobi diverged on all of them, and damped Jacobi and steepest descent exhausted the 20,000-iteration budget. Gauss–Seidel converged only on two of the coarsest meshes: with the larger condition numbers of these problems it becomes impractical.

6.5 Validation and remaining limits

Across the four refinement references, prescribed displacements are exactly zero and true relative residuals are below 1.3×10^{-13}. Normalized force and moment imbalances are below 4×10^{-14}, counting reactions only at constrained components. Force imbalance is normalized by the sum of applied nodal force magnitudes; moment imbalance about the origin uses that force scale times the square's diagonal length. The separate roller-supported affine test gives a relative nodal displacement error of 2.12×10^{-15} against its exact field; it does not provide an exact answer for the clamped benchmark.

The evidence therefore separates algebraic solution accuracy, mesh dependence and verification against a known solution. It does not establish continuum accuracy or physical validity near the clamp corners and the corners of the opening, where stresses concentrate. No material or thickness comparison is made.

References

[1] NLA project team, *NLA: force-driven square and solver benchmark*, `https://github.com/Pueblo98/NLA`, 2026, Source snapshot 9736c8e; unchanged implementation files accompany the section's reproducibility package.

[2] Yousef Saad, *Iterative methods for sparse linear systems*, second ed., SIAM, 2003.